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Student Submissions — Class Work (30.7.26)

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Student 25 **Perarasan V** 192521404RD

10 / 10 submitted

Q1. Peak Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
def peak_element(arr):
    n = len(arr)
    for i in range(n):
        left = arr[i-1] if i-1 ≥ 0 else float('-inf')
        right = arr[i+1] if i+1 < n else float('-inf')
        if arr[i] ≥ left and arr[i] ≥ right:
            return arr[i]

arr = list(map(int, input().split()))
print(peak_element(arr))
```

INPUT

1 3 20 4 1 0

OUTPUT

20

Q2. Second Largest and Second Smallest - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
nums = tuple(map(int, input().split()))
unique_nums = sorted(set(nums))
if len(unique_nums) < 2:
    print("Not enough unique elements")
else:
    print(unique_nums[1])
    print(unique_nums[-2])
```

INPUT

10 20 5 40 15 30

OUTPUT

10
30

Q3. Frequency of Each Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
arr = list(map(int, input().split()))

freq = {}
for num in arr:
    if num in freq:
        freq[num] += 1
    else:
        freq[num] = 1

for key in freq:
    print(f"{key} = {freq[key]}")
```

INPUT

1 2 3 2 4 1 5 2

OUTPUT

1 = 2

```
2 = 3
3 = 1
4 = 1
5 = 1
```

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Q4. Missing Number - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
def missing_number(t):
    n = len(t) + 1
    expected_sum = n * (n + 1) // 2
    actual_sum = sum(t)
    return expected_sum - actual_sum

t = tuple(map(int, input().split()))
print(missing_number(t))
```

INPUT

1 2 3 5 6

OUTPUT

4

Q5. Majority Element -List

Score: 10.00 TC: 1/1 passed

CODE

```
def majority_element(arr):
    n = len(arr)
    candidate = None
    count = 0

    for x in arr:
        if count == 0:
            candidate = x
            count = 1
        elif x == candidate:
            count += 1
        else:
            count -= 1

    if arr.count(candidate) > n // 2:
        return candidate
    return None

arr = list(map(int, input().split()))
print(majority_element(arr))
```

INPUT

2 2 1 2 3 2 2

OUTPUT

2

Q6. Sum & Average - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
t = tuple(map(int, input().split()))
s = sum(t)
avg = s / len(t)
print(s)
print(avg)
```

INPUT

10 20 30 40 50

OUTPUT

150
30.0

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Q7. Element Exists -Tuple

Score: 10.00 TC: 2/2 passed

CODE

```
arr = tuple(map(int, input().split()))
x = int(input())
```

```
if x in arr:
    print("Element found")
else:
    print("Element not found")
```

INPUT

```
10 20 30 40 50
25
```

OUTPUT

```
Element not found
```

Q8. Maximum Element - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
nums = tuple(map(int, input().split()))
print(max(nums))
```

INPUT

```
12 45 8 67 23
```

OUTPUT

```
67
```

Q9. Lucky Number

Score: 10.00 TC: 2/2 passed

CODE

```
def digit_sum(n):
    if n < 10:
        return n
    total = 0
    while n > 0:
        total += n % 10
```

```
        n //≠ 10
    return digit_sum(total)

num = int(input())
result = digit_sum(num)

if result == 7:
    print("Lucky Number")
else:
    print("Not a Lucky Number")
```

INPUT

123

OUTPUT

Not a Lucky Number

Q10. Remove Duplicate Elements - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
t = tuple(map(int, input().split()))
result = tuple(set(t))
print(result)
```

INPUT

10 20 30 20 40 10 50

OUTPUT

(40, 10, 50, 20, 30)